Planning Just Bioeconomy Transitions:

Employment Effects of Olive-Tree Pruning Biorefinery Cluster Expansion under Stringent Greenhouse Gas Mitigation Targets

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Introduction

Context: Olive-tree pruning is an abundant agricultural residue whose valorization can contribute to regional decarbonization and rural development.

Multiple transformation routes (biochemical, thermochemical and physicochemical) have been proposed. We evaluate 32 subprocesses, 9 byproducts and **8 products:**

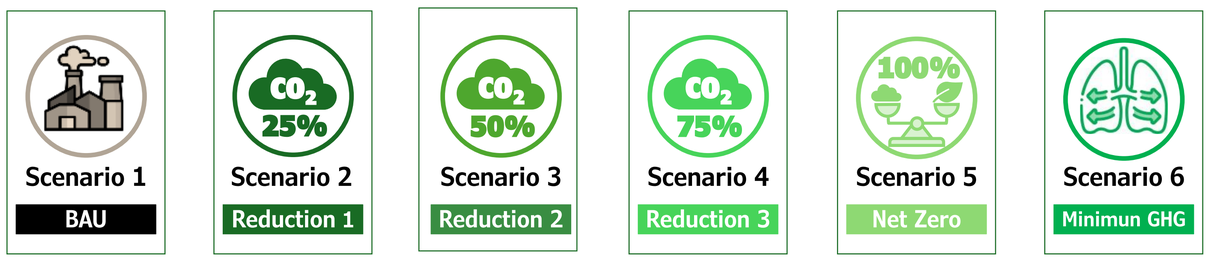


Challenge: Stringent climate targets may substantially change the optimal configuration of biomass valorization networks, with potentially important consequences for employment.

Objective: Assess how increasingly stringent GHG mitigation targets reshape an olive-pruning biorefinery network and quantify the associated direct, indirect and induced employment effects.

Methodology

Optimal transformation networks: Progressive **decarbonization** targets by using LCA factors and actual conversions.



Biomass network superstructure model: Linear a programming optimization framework.



PYOMO python COIN-OR — *Pyomo of Python with the COIN Branch and Cut solver*

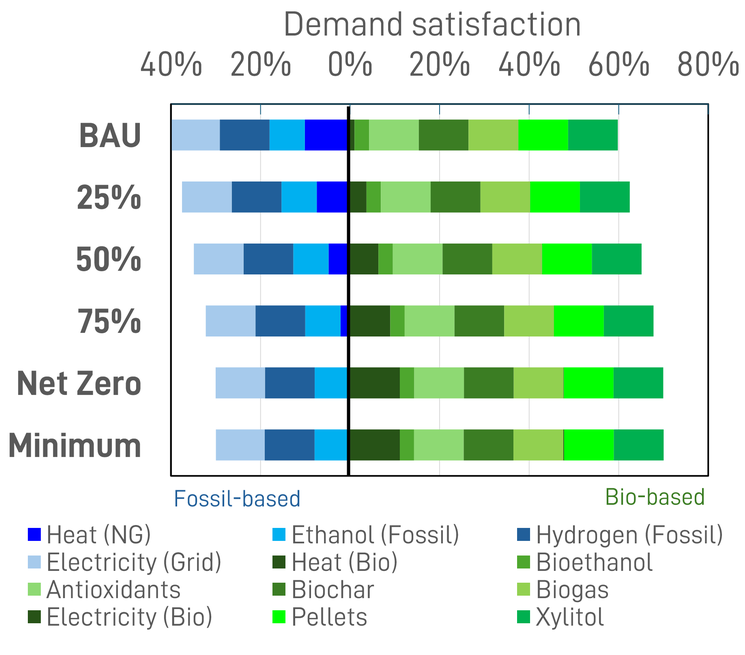
Employment metric: annual full-time equivalent (FTE) jobs, including direct, indirect, and induced employment

I-JEDI

Results & Discussion

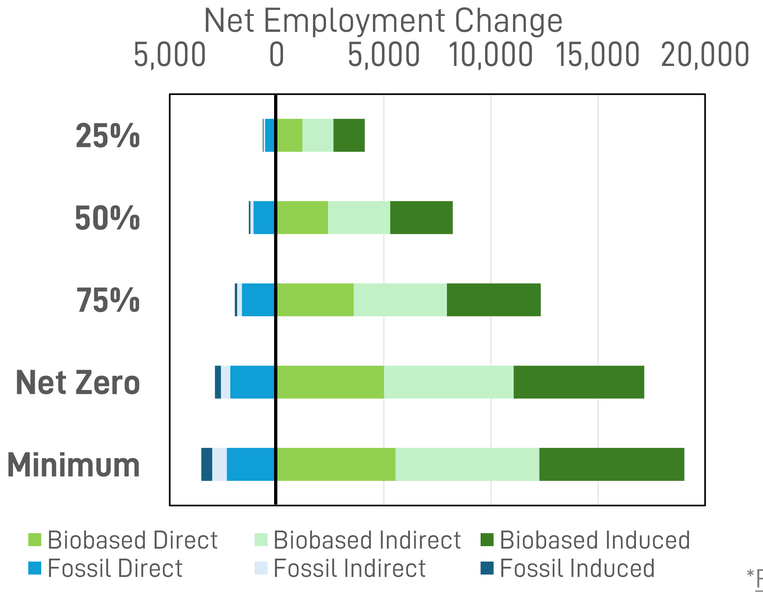
Increasing decarbonization efforts progressively substitute fossil supply with sustainable bio-based production.

a) Main products by Climate Scenario



Decarbonization steadily expands biobased jobs (direct, indirect, and induced), easily outstripping fossil losses across scenarios.

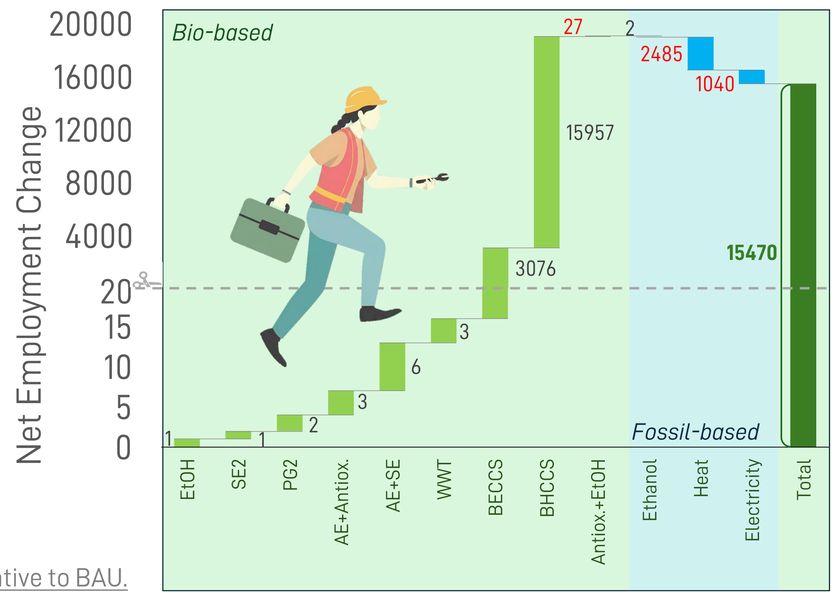
b) Net Employment Change* by Climate Scenario



*Relative to BAU.

Deep decarbonization generates up to ~15.500 net jobs in Minimum GHG scenario relative to BAU .

c) Technology contribution to Net Employment Change* in Minimum Scenario



Conclusions

- The proposed framework allows for **planning bioeconomic transitions** aligned with climate goals and regional development.
- The transition toward low-carbon infrastructure is primarily driven by the **restructuring of ethanol and heat supply**.
- Tighter GHG constraints progressively **reshape the optimal olive-pruning valorization network**.
- Bio-based job creation consistently exceeds fossil-related job losses, yielding **positive net employment effects** across decarbonization scenarios.
- Deep decarbonization produces the **largest employment gains**, particularly through indirect and induced effects.
- Progressive decarbonization drives life-cycle GHG emissions net-negative, reaching roughly **–13 Mt CO_2-eq·yr^{-1} in the minimum—GHG scenario**.

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[2] NREL. *JEDI: Jobs and Economic Development Impact Model*. National Laboratory of the Rockies (**2026**): U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy.

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